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◇ Spring Boot Core (1–25)

1. What is Spring Boot?
2. How is Spring Boot different from Spring Framework?
3. What are the advantages of Spring Boot?
4. What is auto-configuration in Spring Boot?
5. How does Spring Boot simplify dependency management?
6. What is spring-boot-starter?
7. What is Spring Initializr?
8. What is the default embedded server in Spring Boot?
9. How do you change the default port?
10. What is the use of application.properties vs application.yml?
11. What is @SpringBootApplication?
12. What are the components of @SpringBootApplication?
13. What is component scanning?
14. What is the use of @Component, @Service, @Repository?
15. What is dependency injection?
16. What is Inversion of Control (IoC)?
17. What is the Spring Boot lifecycle?
18. How do you run a Spring Boot application?
19. What is CommandLineRunner?
20. What is ApplicationRunner?
21. What is the difference between them?
22. What is Spring Boot DevTools?
23. What is hot reloading?
24. How do you disable auto-configuration?
25. What is @EnableAutoConfiguration?

◇ REST API & Web (26–50)

26. What is REST?
27. What is @RestController?
28. Difference between @Controller and @RestController?
29. What is @RequestMapping?
30. Difference between GET, POST, PUT, DELETE?
31. What is @GetMapping, @PostMapping, etc.?
32. What is path variable vs request param?
33. What is @RequestBody?
34. What is @ResponseBody?
35. How do you validate request payloads?
36. What is @Valid?
37. What is @Validated?
38. What is exception handling in Spring Boot?
39. What is @ControllerAdvice?
40. What is @ExceptionHandler?
41. What is ResponseEntity?
42. What are HTTP status codes?
43. What is content negotiation?
44. What is HATEOAS?
45. How do you handle file uploads?
46. How do you implement pagination?

47. What is sorting in Spring Boot?
48. How do you handle CORS?
49. What is API versioning?
50. What is idempotency?

◇ Spring Data JPA & Hibernate (51–80)

51. What is JPA?
52. What is Hibernate?
53. Difference between JPA and Hibernate?
54. What is an Entity?
55. What is @Entity?
56. What is @Table?
57. What is @Id?
58. What is @GeneratedValue?
59. What are generation strategies?
60. What is a repository?
61. What is JpaRepository?
62. Difference between CrudRepository and JpaRepository?
63. What is query derivation?
64. What is @Query annotation?
65. What is JPQL?
66. What is native query?
67. What is lazy vs eager loading?
68. What is N+1 problem?
69. How do you solve N+1 problem?
70. What is caching in Hibernate?
71. First-level vs second-level cache?
72. What is transaction management?
73. What is @Transactional?
74. Propagation types?
75. Isolation levels?
76. What is dirty checking?
77. What is cascade type?
78. What is orphan removal?
79. What is DTO projection?
80. What is pagination in JPA?

◇ Microservices & Architecture (81–105)

81. What is a microservice?
82. Monolith vs microservices?
83. What are advantages of microservices?
84. What is service discovery?
85. What is Eureka?
86. What is API Gateway?
87. What is Spring Cloud?
88. What is load balancing?
89. What is circuit breaker?
90. What is Resilience4j?
91. What is distributed tracing?
92. What is centralized configuration?
93. What is config server?
94. What is inter-service communication?
95. REST vs messaging?
96. What is Feign Client?
97. What is service registry?

98. What is fault tolerance?
99. What is scalability?
100. What is eventual consistency?
101. What is Saga pattern?
102. What is CQRS?
103. What is event-driven architecture?
104. What is API gateway routing?
105. What is blue-green deployment?

◇ Kafka & Messaging (106–120)

106. What is Apache Kafka?
107. What is a topic?
108. What is a partition?
109. What is a consumer group?
110. What is offset?
111. How does Kafka ensure durability?
112. What is producer acknowledgment?
113. What is message serialization?
114. What is Kafka in Spring Boot?
115. What is @KafkaListener?
116. How do you handle retries?
117. What is dead letter queue?
118. What is event-driven architecture?
119. How do you ensure idempotency in Kafka?
120. What is exactly-once processing?

◇ Testing & Debugging (121–135)

121. What is unit testing?
122. What is integration testing?
123. What is @SpringBootTest?
124. What is @WebMvcTest?
125. What is Mockito?
126. What is @MockBean?
127. What is @InjectMocks?
128. What is TestRestTemplate?
129. What is MockMvc?
130. What is code coverage?
131. How do you test REST APIs?
132. How do you test exception scenarios?
133. What is test slicing?
134. How do you debug Spring Boot apps?
135. What are common debugging tools?

◇ DevOps, CI/CD & Deployment (136–150)

136. How do you deploy Spring Boot apps?
137. What is a fat JAR?
138. What is Docker?
139. How do you containerize Spring Boot apps?
140. What is Kubernetes/OpenShift?
141. What is CI/CD?
142. What is Jenkins pipeline?
143. What is GitLab CI?
144. What is environment configuration?
145. What is profile in Spring Boot?

146. What is @Profile?
147. What is logging in Spring Boot?
148. What is Actuator?
149. What are health checks?
150. How do you monitor Spring Boot applications?